THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2:3	A:B:C $= 6:8:9$
2	**(d)** 24	Total work $= 12 \times 8 \times 15 = 1440$ man-hours
3	**(a)** 720	One part $= 1350 \div 15 = 90$
4	**(a)** 30	Quantity × price is constant
5	**(b)** 12	Distance $= 540$ km
6	**(a)** 32	One part $= 8$; difference $= (11-7) \times 8$
7	**(a)** 32	$80/360 \times 144$
8	**(a)** 4:5	A:B:C $= 4:14:5$
9	**(d)** 18	Total work $= 9 \times 6 \times 20 = 1080$ man-hours
10	**(a)** 480	One part $= 780 \div 13 = 60$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One part $= 143 \div 13 = 11$ (1); numbers 55 and 88 (1).	1+1
12	Student-days $= 50 \times 16 = 800$ (1); days $= 800 \div 40 = 20$ (1).	1+1
13	Students $= 120/360 \times 108 = 36$ (1); percent $= 33.33\%$ (1).	1+1
14	One part $= 160 \div 16 = 10$ (1); numbers 70 and 90 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total work $= 10 \times 6 \times 24 = 1440$ man-hours (1); workers' hours per day $= 12 \times 8 = 96$ (1); days $= 15$ (1).	1+1+1
16	A:B:C $= 3:6:8$ (1); one part $= 10$ (1); A = 30, B = 60, C = 80 (1).	1+1+1
17	Students per degree $= 15/60$ (1); total $= 15 \times 360/60 = 90$ (1); 96° sector $= 24$ students (1).	1+1+1
	OR: Total work $= 15 \times 8 \times 12 = 1440$ man-hours (1); workers' hours per day $= 18 \times 5 = 90$ (1); days $= 16$ (1).	1+1+1
18	A:B:C $= 2:3:5$ (1); one part $= 20$ (1); A = 40, B = 60, C = 100 (1).	1+1+1
	OR: Students per degree $= 12/45$ (1); total $= 12 \times 360/45 = 96$ (1); 120° sector $= 32$ students (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Milk $= 24$ L, water $= 36$ L (2); new water $= 48$ L (1); water to add $= 12$ L (2).	2+1+2
	OR: Work done in 8 days $= 8/20$ of the task; remaining $= \frac{3}{5}$ (1); workers now $= 18$; remaining man-days $= 144$ (2); days $= (12 \times 12) \div 18 = 8$ (2).	1+2+2
20	Milk $= 36$ L, water $= 48$ L (2); new water $= 72$ L (1); water to add $= 24$ L (2).	2+1+2
	OR: Work done in 12 days $= 12/30$ of the task; remaining $= \frac{3}{5}$ (1); workers now $= 18$; remaining man-days $= 180$ (2); days $= (18 \times 10) \div 18 = 10$ (2).	1+2+2

Mistakes to avoid:

- Treating inverse proportion as direct: more workers means **fewer** days, so use $x_1 y_1 = x_2 y_2$.
- Not converting the map scale to the same units (cm to km) before computing.
- Dividing instead of multiplying (or the other way) when converting map distance to actual distance.
- Computing the pie-chart angle as $\frac{\text{value}}{\text{total}} \times 100$ (percentage) instead of $\times 360°$.
- Using direct proportion for quantities that are not proportional (e.g. age and height).
- Not checking that all sector angles add up to $360°$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 24	Quantity × price is constant
2	**(d)** 5	Distance $= 480$ km
3	**(a)** 28	One part $= 7$; difference $= (9-5)\times 7$
4	**(a)** 35	$140/360 \times 90$
5	**(c)** 15:8	A:B:C $= 15:18:8$
6	**(c)** 12	Total work $= 16\times 5\times 18 = 1440$ man-hours
7	**(b)** 400	One part $= 1120 \div 14 = 80$
8	**(b)** 60	Quantity × price is constant
9	**(d)** 4	Distance $= 360$ km
10	**(b)** 60	One part $= 12$; difference $= (8-3)\times 12$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Student-days $= 60\times 14 = 840$ (1); days $= 840 \div 40 = 21$ (1).	1+1
12	Students $= 72/360 \times 180 = 36$ (1); percent $= 20\%$ (1).	1+1
13	One part $= 225 \div 15 = 15$ (1); numbers 60 and 165 (1).	1+1
14	Student-days $= 45\times 16 = 720$ (1); days $= 720 \div 30 = 24$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total work $= 20\times 7\times 9 = 1260$ man-hours (1); workers' hours per day $= 14\times 6 = 84$ (1); days $= 15$ (1).	1+1+1
16	A:B:C $= 3:4:10$ (1); one part $= 15$ (1); A = 45, B = 60, C = 150 (1).	1+1+1
17	Students per degree $= 20/72$ (1); total $= 20\times 360/72 = 100$ (1); 54° sector $= 15$ students (1).	1+1+1
	OR: Total work $= 24\times 5\times 14 = 1680$ man-hours (1); workers' hours per day $= 20\times 7 = 140$ (1); days $= 12$ (1).	1+1+1
18	A:B:C $= 4:5:6$ (1); one part $= 12$ (1); A = 48, B = 60, C = 72 (1).	1+1+1
	OR: Students per degree $= 24/80$ (1); total $= 24\times 360/80 = 108$ (1); 100° sector $= 30$ students (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Milk $= 40$ L, water $= 50$ L (2); new water $= 90$ L (1); water to add $= 40$ L (2).	2+1+2
	OR: Work done in 9 days $= 9/24$ of the task; remaining $= \frac{5}{8}$ (1); workers now $= 25$; remaining man-days $= 225$ (2); days $= (15\times 15)\div 25 = 9$ (2).	1+2+2
20	Milk $= 50$ L, water $= 60$ L (2); new water $= 70$ L (1); water to add $= 10$ L (2).	2+1+2
	OR: Work done in 6 days $= 6/18$ of the task; remaining $= \frac{2}{3}$ (1); workers now $= 30$; remaining man-days $= 240$ (2); days $= (12\times 20)\div 30 = 8$ (2).	1+2+2

Mistakes to avoid:

- Treating inverse proportion as direct: more workers means **fewer** days, so use $x_1y_1 = x_2y_2$.
- Not converting the map scale to the same units (cm to km) before computing.
- Dividing instead of multiplying (or the other way) when converting map distance to actual distance.
- Computing the pie-chart angle as $\frac{\text{value}}{\text{total}} \times 100$ (percentage) instead of $\times 360°$.
- Using direct proportion for quantities that are not proportional (e.g. age and height).
- Not checking that all sector angles add up to $360°$.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 60	$100/360 \times 216$
2	**(b)** 2:15	A:B:C $= 4:9:30$
3	**(a)** 20	Total work $= 8 \times 9 \times 25 = 1800$ man-hours
4	**(d)** 630	One part $= 990 \div 11 = 90$
5	**(b)** 35	Quantity × price is constant
6	**(d)** 12	Distance $= 720$ km
7	**(d)** 90	One part $= 10$; difference $= (13-4) \times 10$
8	**(a)** 21	$60/360 \times 126$
9	**(c)** 21:10	A:B:C $= 21:24:10$
10	**(c)** 28	Total work $= 14 \times 8 \times 21 = 2352$ man-hours

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Students $= 135/360 \times 160 = 60$ (1); percent $= 37.5\%$ (1).	1+1
12	One part $= 190 \div 19 = 10$ (1); numbers 60 and 130 (1).	1+1
13	Student-days $= 72 \times 15 = 1080$ (1); days $= 1080 \div 60 = 18$ (1).	1+1
14	Students $= 150/360 \times 96 = 40$ (1); percent $= 41.67\%$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total work $= 9 \times 10 \times 16 = 1440$ man-hours (1); workers' hours per day $= 12 \times 6 = 72$ (1); days $= 20$ (1).	1+1+1
16	A:B:C $= 2:6:9$ (1); one part $= 10$ (1); A = 20, B = 60, C = 90 (1).	1+1+1
17	Students per degree $= 9/30$ (1); total $= 9 \times 360/30 = 108$ (1); 150° sector $= 45$ students (1).	1+1+1
	OR: Total work $= 18 \times 6 \times 25 = 2700$ man-hours (1); workers' hours per day $= 15 \times 9 = 135$ (1); days $= 20$ (1).	1+1+1
18	A:B:C $= 5:6:8$ (1); one part $= 10$ (1); A = 50, B = 60, C = 80 (1).	1+1+1
	OR: Students per degree $= 7/36$ (1); total $= 7 \times 360/36 = 70$ (1); 72° sector $= 14$ students (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Milk $= 12$ L, water $= 60$ L (2); new water $= 96$ L (1); water to add $= 36$ L (2).	2+1+2
	OR: Work done in 12 days $= 12/36$ of the task; remaining $= \frac{2}{3}$ (1); workers now $= 12$; remaining man-days $= 192$ (2); days $= (24 \times 8) \div 12 = 16$ (2).	1+2+2
20	Milk $= 30$ L, water $= 45$ L (2); new water $= 75$ L (1); water to add $= 30$ L (2).	2+1+2
	OR: Work done in 10 days $= 10/25$ of the task; remaining $= \frac{3}{5}$ (1); workers now $= 30$; remaining man-days $= 270$ (2); days $= (15 \times 18) \div 30 = 9$ (2).	1+2+2

Mistakes to avoid:

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